UNIT-V Troubleshooting and Preventive Maintenance

5.1 POST (power-on self-test)

A power-on self-test (POST) is a process performed by BIOS after a computer is powered on.

When power is turned on, POST (Power-On Self-Test) is the diagnostic testing sequence that a computer's basic input/output system (or "starting program") runs to determine if the computer keyboard, random access memory, disk drives, and other hardware are working correctly. If the necessary hardware is detected and found to be operating properly, the computer begins to boot. If the hardware is not detected or is found not to be operating properly, the BIOS issue an error message which may be text on the display screen and/or a series of coded beeps, depending on the nature of the problem.

POST protects the bootstrapped code from being interrupted by faulty hardware. When problem, found during post, Bios will use three methods to represent problem:

Beep code

Error code

Detail error message

At end all test complete the post passes control to the bootstrap program in Bios.

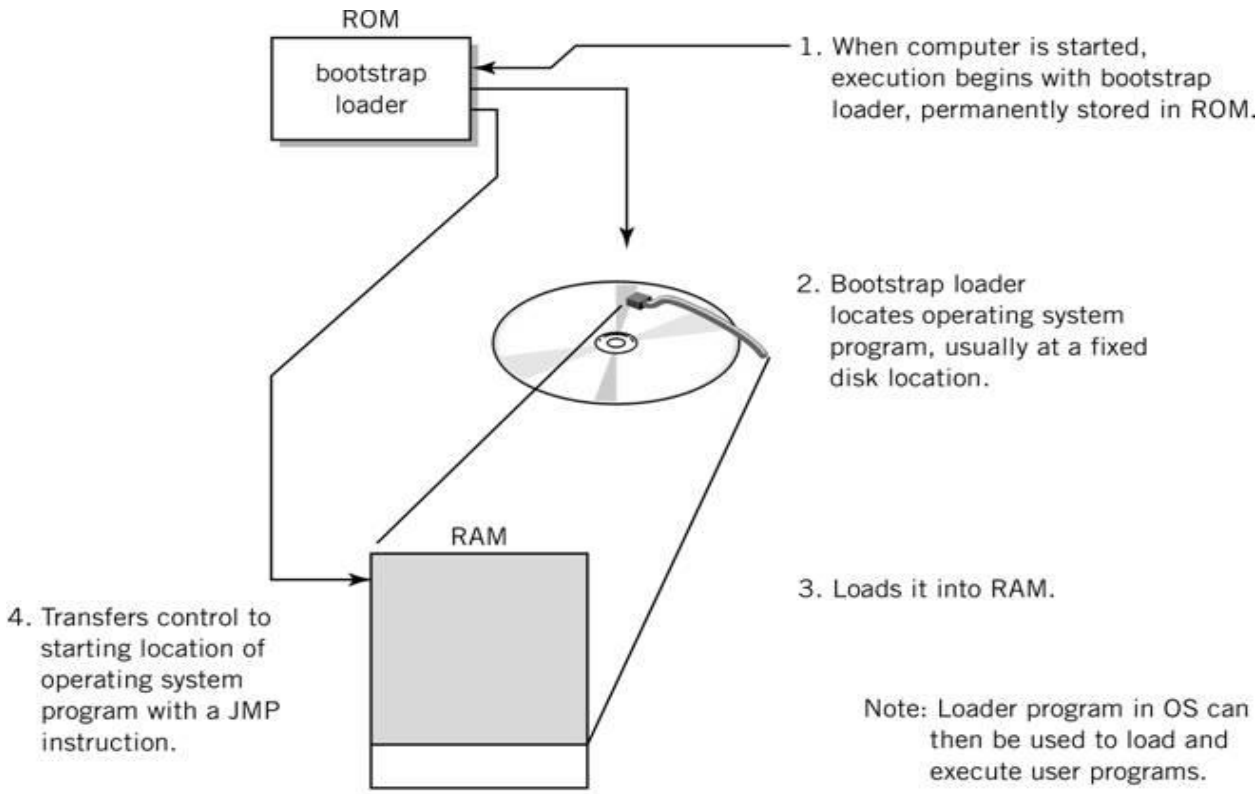
5.1.2 IPL (initial program load) Hardware

IPL Working at booting

- BIOS first job is to check the hardware. This is referred to as the Power-On Self-Test (POST).
- Next, the BIOS must start loading the Operating System from disk, which takes over running the computer. This phase is called the Initial Program Load (IPL).

‘Boot’ code

On PC systems the mechanism for doing the IPL (‘Initial Program Load’) requires a ‘boot-loader’ program that can fit within a single 512-byte disk-sector on ROM called bootstrap loader.



The system initialization process will be referred to as the IPL. IPL (initial program load) is a term for the loading of the operating system into the computer's main memory . On a personal computer, booting or re-booting (re-starting) is the equivalent to IPL Process.

5.1.3 Test Sequence

The computer POST (Power On Self Test) tests the computer, insuring that it meets the necessary system requirements and that all hardware is working properly before starting the remainder of the boot process. If the computer passes the POST the computer will have a single beep (with some computer BIOS manufacturers it may beep twice) as the computer starts and the computer will continue to start normally. However, if the computer fails the POST, the computer will either not beep at all or will generate a beep code, which tells the user the source of the problem.

The steps of a POST

Each time the computer boots up the computer must past the POST. Below is the common step a POST performs each time your computer starts.

1. Test the power supply to ensure that it is turned on and that it releases its reset signal.
2. CPU must exit the reset status mode and thereafter be able to execute instructions.
3. BIOS checksum must be valid, meaning that it must be readable.
4. CMOS checksum must be valid, meaning that it must be readable.
5. CPU must be able to read all forms of memory such as the memory controller, memory bus, and memory module.
6. The first 64KB of memory must be operational and have the capability to be read and written to and from, and capable of containing the POST code.
7. I/O bus / controller must be accessible.
8. I/O bus must be able to write / read from the video subsystem and be able to read all video RAM.

5.1.4 POST error messages

The error messages and codes include all messages generated by servers during the Power On Self Test (POST). Some messages are informational only and do not indicate any error. All POST Error messages and beep codes occur before an operating system has loaded. Therefore, a server generates only the codes that are applicable to its BIOS configuration and hardware options that are installed.

To read the Post error message on screen the video sub-system of the server must be working. In the event that the video is not working, use the beep codes to troubleshoot the cause of the failing video during the Power On Self Test of the server.

Both Macs and Windows PCs run a POST each time the computer is booted up or restarted. The scan checks the hardware and makes sure the processor, RAM, and storage devices are all functioning correctly. If an error is encountered during the POST, the startup process may pause or halt completely and the error may be displayed on the monitor. On PCs, POST errors are often displayed on the BIOS information screen. They may be output as cryptic codes, such as "08" for bad memory, or as a system message, such as "System RAM failed at offset."

Each BIOS manufacturer has its own diagnostic codes that identify specific POST errors. You need to consult the BIOS documentation for the diagnostic codes for your BIOS, but the general breakdown of the code categories is as follows:

- **100–199:** Motherboard error
- **200–299:** Memory error
- **300–399:** Keyboard error
- **600–699:** Floppy drive error
- **1400–1499:** Printer error
- **1700–1799:** Hard drive error

AMI BIOS beep codes

Below are the AMI BIOS Beep codes that can occur. However, because of the wide variety of different computer manufacturers with this BIOS, the beep codes may vary.

Beep Code	Descriptions
1 short	DRAM refresh failure
2 short	Parity circuit failure
3 short	Base 64K RAM failure
4 short	System timer failure
5 short	Process failure
6 short	Keyboard controller Gate A20 error
7 short	Virtual mode exception error
8 short	Display memory Read/Write test failure
9 short	ROM BIOS checksum failure
10 short	CMOS shutdown Read/Write error
11 short	Cache Memory error
1 long, 3 short	Conventional/Extended memory failure
1 long, 8 short	Display/Retrace test failed

AWARD BIOS beep codes

Below are Award BIOS Beep codes that can occur. However, because of the wide variety of different computer manufacturers with this BIOS, the beep codes may vary.

Beep Code	**Description**
1 long, 2 short	Indicates a video error has occurred and the BIOS cannot initialize the video screen to display any additional information
Any other beep(s)	RAM problem.

If any other correctable hardware issues, the BIOS will display a message.

IBM BIOS beep codes

Below are IBM BIOS Beep codes that can occur. However, because of the wide variety of models shipping with this BIOS, the beep codes may vary.

Beep Code	**Description**
No Beeps	No Power, Loose Card, or Short.
1 Short Beep	Normal POST, computer is ok.
2 Short Beep	POST error, review screen for error code.
Continuous Beep	No Power, Loose Card, or Short.
Repeating Short Beep	No Power, Loose Card, or Short.
One Long and one Short Beep	Motherboard issue.
One Long and Two Short Beeps	Video (Mono/CGA Display Circuitry) issue.
One Long and Three Short Beeps.	Video (EGA) Display Circuitry.
Three Long Beeps	Keyboard / Keyboard card error.
One Beep, Blank or Incorrect Display	Video Display Circuitry.

5.2 Troubleshooting: possible problems and diagnosis

Troubleshooting is the process of diagnosing the source of a problem. It is used to fix problems with hardware, software, and many other products. The basic theory of troubleshooting is that you start with the most general (and often most obvious) possible problems, and then narrow it down to more specific issues.

Troubleshooting is a systematic approach to solving a problem. Computer troubleshooting may involve hardware or software and can sometimes involve both at the same time. The goal of troubleshooting is to determine why something does not work as expected and explain how to resolve the problem.

5.2.1Motherboard

The motherboard and processor are two of the most important hardware components inside the computer. The various pieces of hardware inside the PC communicate with one another through the circuits on the motherboard, while the CPU stores and executes programming instructions. The motherboard and CPU can both be expensive to replace, however, but diagnosing hardware failure on your own can reduce the potential repair costs. Hardware components exhibit symptoms when failing.

Problem1: System has no power at all. Power light does not illuminate, fan inside the power supply does not turn on, and indicator light on keyboard does not turn on.

Cause: Power cable is unplugged.

Solution: Make sure power cable is securely plugged in.

Cause: Defective power cable.

Solution: Replace cable.

Cause: Power supply failure.

Solution: Power cable and wall socket are OK, but system is still dead. Replace cable.

Cause: Faulty wall outlet; circuit breaker or fuse blown.

Solution: Use different socket, repair outlet, reset circuit breaker or replace fuse.

Problem 2: System inoperative. Keyboard lights are on, power indicator lights are lit, and hard drive is spinning.

Cause: Expansion card is partially dislodged from expansion slot on the motherboard.

Solution: Turn off computer. Take cover off system unit. Check all expansion cards to ensure they are securely seated in slots. Using even pressure on both ends of the expansion card, press down firmly on expansion card.

Cause: Defective expansion card.

Solution: Turncomputeroff.Remove an expansion card. Make sure expansion card is secure in expansion socket.

Problem 3: System only boots from Floppy Disk. Hard Disk can be read and applications can be used, but booting from Hard Disk is impossible.

Cause: Hard Disk boot program has been destroyed.

Solution: Back up data and applications files. Reformat the Hard Drive as described in the Hard Drive section of the manual. Re-install applications and data using backup disks.

Problem 4: Screen message says "Invalid Configuration" or "CMOS Failure."

Cause: Incorrect information entered into the configuration (setup) program.

Solution: Check the configuration program. Replace any incorrect information.

Make sure correct information is in setup.

Problem 5: Screen is blank.

Cause: No power to monitor. Power connectors may be loose or not plugged in.

Solution: Check the power connectors to monitor and to system. Make sure monitor is connected to display card, change I/O address on network card if applicable.

Problem 6: System does not boot from hard disk drive, can be booted from floppy disk drive.

Cause: Connector between hard drive and system board unplugged.

Solution: Check cable running form disk to disk controller on the board. Make sure both ends are securely plugged in; check the drive type in the Standard CMOS Setup

Problem 7: Floppy drive lights stays on.

Cause: Floppy Drive cable not connected correctly.

Solution: Reconnect floppy cable making sure PIN1 on the Floppy Drive corresponds with PIN1 on floppy cable connector.

Problem 8: C: drive failure.

Cause: SETUP program does not have correct information.

Solution: Boot from drive A: using DOS system disk. Input correct information to SETUP program.

Cause: Hard Drive cable not connected properly.

Solution: Check Hard drive cable.

Problem 9: Cannot boot system after installing second hard drive.

Cause: Master/Slave jumpers not set correctly.

Solution: Set master /Slave jumpers correctly.

Problem 10: Missing operating system on hard drive.

Cause: CMOS setup has been changed.

Solution: Run setup and select correct drive type.

5.2.2 Keyboard

If your keyboard suddenly does not work at all, or certain keys do not seem to be doing what they should, you cannot perform the tasks you need to do. Fortunately, there are several troubleshooting steps you can take to resolve these problems and fix your keyboard.

1. The keyboard is not working.

Possible Causes and Solutions:

- Make sure the keyboard is plugged into the computer correctly and the keyboard cable was inserted while the computer was turned on the computer might not have recognized the keyboard.
- If you have a wireless keyboard, the first troubleshooting step to take is to try installing fresh batteries. Also, make sure the adapter for the wireless keyboard is inserted properly into the computer.
- If the keyboard is still unresponsive after trying these troubleshooting steps, check the status of keyboard's driver. Update Driver.
- If it is still unresponsive, try installing it onto another computer, if possible. If the keyboard does not work at all on a different machine, then the keyboard itself is damaged and will need to be replaced.

2. The computer indicates “Keyboard error” during start-up or booting.

Possible Causes and Solutions:

- **The keyboard is not plugged properly.**

  Turn-off the power, plugs the keyboard into its port, and reboot. If it is plugged in, try another keyboard in its place.

- **The Keyboard interface is incorrect.**

The keyboard can have the wrong interface as a setting, please reset the keyboard.

3. The keyboard output is very slow or some characters are lost when typing

Possible Causes and Solutions:

- Wrong configuration.
- Change the output speed of the keyboard.

4. Keyboard letters don't type.

Possible Causes and Solutions:

Sometimes a tiny piece of garbage can get stuck under a few of your keys. If you find that a certain letter won't type, you may be able to fix the problem by using a compressed air duster and gently blowing off your keys.

5. Keyboard numbers don't type.

Possible Causes and Solutions:

There is a "numbers lock" button near your keypad that turns the pad on and off. If your numbers won’t type, you’ve probably pressed this button by mistake. so on the numlock key.

6. Windows doesn't recognize keyboard, or doesn't recognize some keys.

Possible Causes and Solutions:

The driver for your keyboard might be missing or need to be updated. So update driver.

7. Some keys stick or don't depress properly.

Possible Causes and Solutions:

Cause: Keyboards get very dirty sometimes.
Solution: You can clean a keyboard.

8. The characters I type don't match the ones that show up on the monitor.

Possible Causes and Solutions:

- You press on the keyboard do not match the characters that are displayed on the computer.
- If so, your computer may be using the incorrect language settings. To troubleshoot and fix this problem correct language and country or region are selected.

9. SHIFT, CTRL, ALT, or Windows Logo key isn't working.

Possible Causes and Solutions:

You might have activated the Sticky Keys accessibility feature.

10. Keyboard is locked, no keys function.

Possible Causes and Solutions:

Keyboard is locked. Unlock keyboard.

11. Keyboard Not Found" Message.

Possible Causes and Solutions:

Your keyboard is not plugged into the computer securely. Unplug it and plug it back in and the problem should go away.

12. Computer isn't taking inputs from Keyboard.

Possible Causes and Solutions:

- Is keyboard connected to computer? Ensure that the keyboard is plugged into the keyboard jack and not into the mouse jack. If the keyboard was unplugged, plug it back in and reboot the computer.
- If the keyboard still doesn't work on boot-up, power down the computer .Plug the new keyboard up and boot up the computer. If the new keyboard works, the old keyboard is bad and needs to be replaced.
- If the new keyboard doesn't work, check your BIOS to make sure it sees the keyboard. If the BIOS recognize the keyboard, then you probably have a bad keyboard port.

13. Keys Don't Work.

Possible Causes and Solutions:

- If one of the keys on your keyboard gets stuck, turn the computer off and clean with a damp cloth.
- If only some of the keys on your keyboard are not working, The first would be to check the device status and update the keyboard's driver.

14. Certain keys do not function.

Possible Causes and Solutions:

Keys jammed or defective. Replace keyboard.

5.2.3Hard Disk Drive

Hard disks are often containing the bundle of important information and a huge capacity to store valuable data. But no one would take interest to backup their data or important files so that it can be recovered anytime at the time of failure or corruption. Hard disk failure can happen anytime due to bad power supply, virus attack, damaged block and sectors, corrupted operating system files.

1. Corrupted files on hard disk

Sometimes, the files become inaccessible by the user as the operating system fails to boot up properly. The reasons for this could be shutting down the device unethically, closing the running program accidentally, power failure, use of corrupted software etc. All these things could potentially contribute a lot in corrupting the hard disk files.

Solution: – Always shut down the system by following a specific way. Avoid installing suspected software and malicious programs into your hard disk. Clean the hard disk on a regular basis to remove unwanted programs.

2. Virus threat

Virus attack is one of the major problems that are faced by HDD. This problem occurs when the hard disk is attacked by a virus or malware program. It even spreads to other systems across the network. Your system may get virus from other media, external drives, network etc. If the user takes back-up of files that contain virus, then the system gets re-infected and results in hard disk failure.

Solution: – The best solution for this problem is updating the system on a regular basis with the help of a reliable anti-virus program. It is necessary to update the anti-virus program frequently so that it can detect almost all kinds of viruses and malware programs.

3. Unexpected computer crashes

This may occur when the hard disk has too many bad sectors and blocks. Sometimes, the hard disk may crash as the spindle motor stops rotating. You will notice symptoms like – clicking sounds from HDD, invisible files and folders etc. The heads may not be able to identify the sectors the user is searching for hence, are making contact with the platters. This problem occurs when the hard disk is formatted more than its limit or is too old to operate.

Solution: – To get rid of this problem, users must update their system on a regular basis by using a reliable software program. It is recommended to replace the HDD after 3-4 years of usage to avoid this problem.

4. Computer fails to detect hard disk or BIOS

When the UPS fails to deliver the required power supply or gives very high power supply, the BIOS cannot detect the hard drive. The HDD fails to spin up in such situations. You may even have problem with the mother board.

Solution: – To avoid this problem, make sure that your system is getting adequate power supply. You can troubleshoot this problem by using UPS of a reputed brand. A good course of action is to replace the IDE, ATA, or SCSI cable. You can even switch the drive to another power plug.

5. Manufacturing fault

This kind of problem occurs with a new hard disk when it is delivered to the customer without testing. The hard disk may work well initially, but after a few months, it becomes non-responsive.

Solution: - The solution to this problem is proper testing. Buyers should get the hard disk tested before bringing it home. In case you face any problem after installing it, it should be replaced.

6. Heat and dust

This is one of the hard disk problems that occur frequently. Due to excess usage, the system gets heated to the peak point. Sometimes, lack of ventilation and faulty CPU fan also cause this problem.

Solution: – Make sure that the CPU fans are in proper condition. Check whether the processor is getting sufficient cooling or not. Users can make use of SMART tools that notify them about the condition of the hard disk and risks that the hard disk may face in future.

Heat

Symptoms:

- Fans are moving too slowly or not at all.
- Clicking sounds or noise is arising from the system hardware.

Reason:

This is the most common problem among all the hard disk failure causes that occur

too frequently. Due to off improper ventilation or faulty CPU fan, the user system would heat to peak point and leads to hard disk crash or severe damage to the hardware.

Solution:
Ensure that the CPU fans are working fine and give sufficient cooling to the processor and other peripherals. Using S.M.A.R.T. (Self monitoring, analysis and reporting technology) tools will notify the user about the hard disk conditions and any future risks associated with the hardware.

Electronic Failure or Power Surges
Symptoms:

- System BIOS will unable to detect the hard drive,
- Hard drive couldn't be able to spin up.

Reasons:
The UPS that were used by the user couldn't be able to deliver the required power supply or deliver too high power supply to the system.

Solution:
Make sure that the system hardware is getting only the required power supply. By using UPS of any well known brand could troubleshoot this.

Mechanical or Internal failure:

This error happens internally into the hard disk due to of bad sectors and blocks, spindle motor stops to rotate, power supply or any other well known internal factors.

Symptoms:

- Clicking sounds from the hard drive
- Files & Folders become invisible by the time.

Reasons:
The hard drive is too old to operate or formatted more than it's specified limit. Virus

attack can harm the hard disk by erasing the data, altering the hard disk data or system files will also be considered factors for mechanical failure.

Solutions:

Updating the system regularly or weekly, using standard antivirus program, replacing the hard drive after 3-4 years of use. These steps could help to fix the internal errors of the hard drive. Moreover, sometime the hard disk internal errors happen due to unidentified reasons and users are required to have any data recovery software or servicesforcompleterecovery.

Corrupted Files

Symptoms:

- Files are become inaccessible by the user
- Operating system will fail to boot up in a proper manner

Reasons:

Unethical way to shut down the PC, accidentally closing the running program, use of malicious or freeware software, power failures are the factors which contributed the most towards file corruption after which the hard disk become unstable and stops to function properly.

Solution:

Always using the specified way to shut down the PC or closing the program and not installing any freeware or malicious software into your PC will keep the hard disk in running status for long time.

Human Errors

This error happens mainly due to of human mistakes and caused to serious damaging of the system. For ex, modifying the system registry files, altering the system files attribute or any other. These types of errors were hardly recoverable and require the most powerful recovery software to scan deeply and recover seriously damaged data.

1. **The system does not recognize the drive.**

Possible Causes and Solutions:

- Check all cables.
- Make sure the power supply is adequate for system needs.
- Reboot the computer and listen to make sure the drive motor starts up. If the drive is very quiet, it may be difficult to hear its discs reach operating speed. If the drive motor does not start up, recheck all drive cables.
- Verify that for each drive, a drive-type is listed in the system setup program.

- Try rebooting your computer. If the drive is recognized after you reboot the system, the computer BIOS test may be completing before the drive is ready.

2. The system error message, "Drive not ready," appears.

Possible Causes and Solutions:

- Check all cable connections. Make sure pin 1 of the drive is connected to pin 1 of the hard-disc controller or host adapter.
- Make sure the power supply is adequate for system needs.
- Reboot the computer and make sure the drive spins up.

3. The system error message, "HDD controller failure" appears.

Possible Causes and Solutions:

- Confirm the jumper settings on the drive.
- Verify the drive-type settings in the system setup program.

4. Hard drive error occurs.

Cause:

Hard disk has bad sectors or has failed.

Solution:

Select a drive. Under Error-checking, click Check Now. Use a utility to locate and block usage of bad sectors. If necessary, reformat the hard disk.

5. Disk transaction problem.

Cause: Either the directory structure is bad or there is a problem with a file.

Solution: Select a drive. Select File > Properties > Tools. Under Error-checking, click Check Now.

6. Drive not found.

Cause: Cable could be loose.

Solution: Check cable connections.

Cause: The system may not have automatically recognized a newly installed device.

Solution: See if the system still does not recognize the new device, check to see if the device is listed within Computer Setup. If it is listed, the probable cause is a driver problem. If it is not

listed, the probable cause is a hardware problem. If this is a newly installed drive, enter Computer Setup and try adding a POST delay under Advanced > Power-On.

Cause: Drive's controller is disabled in Computer Setup.

Solution: Run Computer Setup and enable all the SATA controllers in Storage > Storage Options.

7. Computer will not start.

Cause: Hard drive is damaged.

Solution: Observe if the front panel Power LED is blinking RED and if any beeps are heard. See "POST Error Messages" to determine possible causes for the blinking red and beep codes.

8. Keyboard lights are on, power indicator lights are lit and hard drive is spinning.

Possible Causes and Solutions:

- Expansion card is partially dislodged from expansion slot on the motherboard. Turn off computer. Take cover off system unit. Check all expansion cards to ensure they are securely seated in slots. Using even pressure on both ends of the expansion card, press down firmly on expansion card.
- Defective expansion card. Turn computer off. Remove an expansion card. Make sure expansion card is secure in expansion socket.

9. System does not boot from hard disk drive can be booted from floppy disk drive.

Possible Causes and Solutions:

- Connector between hard drive and system board unplugged.
- When attempting to run the FDISK utility described in the HARD DISK section of the manual you get a message, INVALID DRIVE SPECIFICATION.
- Check cable running form disk to disk controller on the board. Make sure both ends are securely plugged in;
- Check the drive type in the Standard CMOS Setup.
- Damaged Hard Disk or Disk Controller. Format hard disk; if unable to do so, the hard disk may be defective. Contact Technical Support.

10. After install an expansion card (network card, tape drive card, etc.) the system no longer works properly.

Possible Causes and Solutions:

- No power to monitor. All or part of the system may be inoperable. The new card may work but a mouse or COM port may not work.
- Change the interrupt or RAM address on the new expansion card. See the documentation that came with the new card in order to change pin settings. Many expansion devices come with proprietary software that will assist you in doing this.

11. Screen message says "Invalid Configuration" or "CMOS Failure."

Possible Causes and Solutions:

- Incorrect information entered into the configuration (setup) program.
- Check the configuration program. Replace any incorrect information. Review system's equipment. Make sure correct information is in setup.

12. Screen is blank.

Possible Causes and Solutions:

- No power to monitor. Power connectors may be loose or not plugged in. Monitor not connected to computer. Network card I/O address conflict.
- Check the power connectors to monitor and to system. Make sure monitor is connected to display card, change I/O address on network card if applicable.

13. System does not boot from hard disk drive can be booted from USB drive.

Possible Causes and Solutions:

- Connector between hard drive and system board unplugged.
- When attempting to run the FDISK utility described in the HARD DISK section of the manual you get a message, INVALID DRIVE SPECIFICATION.

- Check cable running form disk to disk controller on the board. Make sure both ends are securely plugged in; check the drive type in the Standard CMOS Setup.

14. Cannot boot system after installing second hard drive.

Possible Causes and Solutions:

- Master/Slave jumpers not set correctly. Set master /Slave jumpers correctly.
- Hard Drives not compatible / different manufacturers.
- Run SETUP program and select correct drive types.
- Call drive manufactures for compatibility with other drives.

15. Missing operating system on hard drive.

Possible Causes and Solutions:

CMOS setup has been changed. Run setup and select correct drive type.

16. C: drive failure.

Possible Causes and Solutions:

- SETUP program does not have correct information. Boot from drive A: using DOS system disk. Input
- Correct information to SETUP program.
- Hard Drive cable not connected properly. Check Hard drive cable.

17. The HDD is not being detected by the BIOS.

Possible Causes and Solutions:

- Check if you connected the IDE cable to the motherboard correctly.
- Check whether the pin 1 on the IDE cable connected to pin 1 on the IDE sockets on both motherboard connector and HDD connector.
- Check if the HDD jumper is set to master and any other device sharing the same cable is set to slave.

Firmware or Manufacturer Faults:

Symptoms:

- The hard disk is undetectable or not recognized at all by the System.
- The system fails to boot or hang in mid of the booting process.

Reason:

This problem could occur mainly with the brand new hard disk after which the hard disk stops to work after few days of purchasing because of manufacturers fault. These types of faulty hard disks are delivered to customer without testing.

Solution:

Users have to contact their hard disk customer center for replacement if possible, therefore, this type of error doesn't require paying extra amount except to lose some data, which is stored in the hard drive before corruption. Manufacturer doesn't guarantee it to recover, so the users are required to have any well known data recovery software for complete recovery of important files and folders from the corrupted or formatted hard drive.

5.2.4Printer

Printers can present a range of problems to consumers. They run through ink too fast, their output quality may be flawed, they may print at a snail's pace, and paper may regularly jam. Fortunately, many of these situations are resolvable.

1. The printer is not working.

Possible causes and solutions:

- Printer is not turned on and online. Turn the printer on and make sure it is online.
- The correct printer drivers for the application are not installed. Install the correct printer driver for the application. If the printer works, reload the printer driver.
- If you are on a network, you may not have made the connection to the printer. Make the proper network connections to the printer.
- Printer may have failed. Run printer self-test.
- It's possible that the driver has become corrupted; it should be reinstalled.

2. Printer running without ink.

Possible causes and solutions:

Printers have some sort of ink tank level indicator, and will issue warnings when your ink is low. You to replace one or more cartridges.

3. An error message about "spooler" problems, or the printer is printing more slowly than usual.

Possible causes and solutions:

If you see an error message about the print spooler, spooler subsystem, or spooler resources, you might need to change and then restart the Print Spooler service on your computer.

4. The printer only prints part of a page, prints blurry ink, or prints faded or inaccurate colors.

Possible causes and solutions:

You might need to replace an ink or toner cartridge. Check the status information in the print queue for low ink or toner messages. If the printer has a status or display area, you can also check for low ink or toner messages on the printer itself.

5. The printer is out of paper.

Possible causes and solutions:

Reload the paper tray or, if the printer has more than one tray, make sure that you reload the tray that contains the paper size you selected. If you add paper but still receive an out-of-paper message, check the print options for the document to see if you've selected the correct paper tray or source.

6. Printer will not turn on.

Possible causes and solutions:

The cables may not be connected properly. Reconnect all cables and check the power cord and electrical outlet.

7. Printer prints corrupted information.

Possible causes and solutions:

- The correct printer driver for the application is not installed. Install the correct printer driver for the application.
- The cables may not be connected properly. Reconnect all cables.
- Printer memory may be overloaded. Reset the printer by turning it off for one minute, and then turn it back on.

8. Printer is offline.

Possible causes and solutions:

The printer may be out of paper. Check the paper tray and refill it if it is empty. Select online.

9. Printer can nothing prints.

Possible causes and solutions:

- Check the power. Make sure the power cord is firmly connected to the printer and to a working outlet, and that the printer is turned on. The Power light on the front panel of the printer should be lit.
- Check the paper. Make sure the paper is loaded correctly and that there is no paper jammed in the printer.
- Check the print cartridges. Make sure that both the black and color print cartridges are properly installed and that the printer's access covers is closed. The Cartridge light will flash if the print cartridges are not installed correctly.

10. A blank page is ejected.

Possible causes and solutions:

- Check that there is no tape covering the ink nozzles on the print cartridges.
- Check that the media being used is wide enough. The media width in the page settings and print settings must match.
- Check for an empty print cartridge. When trying to print black text and a blank page is ejected from the printer, the black print cartridge may be empty. Replace the black

print cartridge. When trying to print using color, and one or more colors do not print properly (or at all), the color cartridge may need to be replaced.

- Check the printer setup. Make sure the correct printer is selected as the current or default printer.
- Check the parallel port on the computer. If a parallel cable is being used, make sure the printer is connected directly to the computer's parallel port. Do not share the port with other devices such as a zip drive.

11. Placement of the text or graphics is wrong.

Possible causes and solutions:

- The paper size or orientation settings may be incorrect. Make sure the paper size and page orientation selected in the software program matches the settings.
- The paper may not be loaded correctly.
- The margin settings may be wrong. If text or graphics are cut off at the edges of the page, make sure the margin settings for the document do not exceed the printable area of the printer.

12. Paper is jammed in the printer

Possible causes and solutions:

- Make sure nothing is blocking the paper path.
- Load paper properly.
- Always use paper that conforms to those listed in the Printer Specifications section of the User's Guide.

13. Printer makes noise during printing.

Possible causes and solutions:

The loud grinding noise during printing is often caused by the carriage rail not having a sufficient amount of lubrication.

14.The printer does not load single-sheet paper or does not feed it properly.

Possible causes and solutions:

Continuous paper is in the paper path. Press the Load/Eject button to feed the continuous paper back to the standby position. Set the paper release lever to the single-sheet position and insert a new sheet.

The paper on the paper guide is positioned too far to the left or right for the printer to detect that it is there. Move the paper (and the edge guides) a little to the left or right.

Paper is creased, old, too thin, or too thick. See Paper for information on paper specifications and use new, clean sheets of paper.

15. Printing appears too high or low on the page.

Possible causes and solutions:

The paper size setting in your application software or printer driver does not match the size of the paper you are using. Check the paper size setting in your application or printer driver.

The printer driver settings you want to use are not selected in your software. Choose the correct printer driver setting from the Windows desktop or in your application before printing.

The printer is on but it doesn't print.	
Cause	**What to do**
The interface cable is broken or twisted.	Check that the interface cable is not broken or twisted. If you have another interface cable, replace the cable and check whether the printer works correctly.
The printer is not connected directly	When you use a printer changer, printer buffer,

to your computer.	or a extension cable, the printer does not print correctly due to the combination of these devices. Make sure that the printer works correctly when it is connected to your computer without these devices.
The printer driver is not installed correctly.	Check that the printer driver is installed properly. If necessary, reinstall the printer driver.
Paper is jammed or the ribbon or guard mechanism is stuck.	Turn the printer off, open the printer cover and remove the jammed paper or things that are stuck.
The USB setting is incorrect.	Confirm the USB setting is correct. See Correcting USB Problems for details.
The printer is hung-up (stopping in abnormal situations).	Turn the printer off and wait for a while. Then, turn the printer on again, and start printing.
The printer does not print when the computer sends data.	
Cause	**What to do**
The printer is paused.	Press the **Pause** button to turn off the Pause light.
The interface cable is not plugged in securely.	Check that both ends of the cable between the printer and the computer are plugged in securely. If the cable is connected correctly, run a self test as described in Printing a Self Test.
Your interface cable does not meet the printer and/or computer specifications.	Use an interface cable that meets the requirements of both the printer and computer. See Electronic and your computer documentation.
Your software is not properly set up for your printer.	Select your printer from the Windows desktop or in your application. If necessary, install or reinstall the printer driver software as described in the Setup Sheet.

The printer makes noise, but nothing prints or printing stops abruptly.	
Cause	**What to do**
The paper-thickness lever is set incorrectly.	Set the paper-thickness lever to match the thickness of your paper.
The ribbon has too much slack, the ribbon has come loose, or the ribbon cartridge is not installed properly.	Remove any slack in the ribbon by turning the ribbon-tightening knob, or reinstall the ribbon cartridge as described in the section under Replacing the ribbon cartridge.
The printer does not load single sheets of paper or does not feed paper properly. (Two or more sheets of paper are fed at the same time.)	
Cause	**What to do**
The paper is jammed together with static electricity.	Fan the paper well before you use it.
The printout is light.	
Cause	**What to do**
The printing speed is set to Ultra high speed draft (USD).	Change the printing speed setting as described in Selecting a Font and Pitch.
The ribbon is damaged out.	Replace the ribbon cartridge as described in Replacing the ribbon cartridge.
The printout is weak.	
Cause	**What to do**
The paper-thickness lever is set incorrectly for your paper.	Move the paper-thickness lever to a lower setting.
The ribbon is damaged out.	Replace the ribbon cartridge as described in Replacing the ribbon cartridge.
Part of the printout is missing, or dots are missing in random positions.	
Cause	**What to do**
The ribbon has too much slack, the ribbon has come loose, or the ribbon cartridge is not installed properly.	Remove any slack in the ribbon by turning the ribbon-tightening knob, or reinstall the ribbon cartridge as described in Replacing the ribbon

	cartridge.
The ribbon is damaged out.	Replace the ribbon cartridge as described in Replacing the ribbon cartridge.
Lines of dots are missing in the printout.	
Cause	**What to do**
The print head is damaged.	Stop printing and contact your dealer to have the print head replaced.
All the text is printed on the same line.	
Cause	**What to do**
No line feed command is sent at the end of each line of text.	Turn on the auto line feed setting using the printer's default-setting mode so that the printer automatically adds a line feed code to each carriage return.
The printout has extra blank lines between lines of text.	
Cause	**What to do**
Two line feed commands are being sent at the end of each line of text, because the printer's auto line feed setting is on.	Turn off the auto line feed setting using the printer's default-setting mode.
The line space setting in your application software is incorrect.	Adjust the line space setting in your application software.
The printer prints strange symbols, incorrect fonts, or other inappropriate characters.	
Cause	**What to do**
Your printer and computer are not communicating correctly.	Make sure that you use the correct interface cable and that the communications protocol is correct.
The wrong character table is selected in your software.	Select the correct character table using your software or the default-setting mode. The character table setting in your software

	overrides settings made in the default-setting mode.
The printer you want to use is not selected in your software	Choose the correct printer from the Windows desktop or in your application before printing.
Your application software is not configured correctly for your printer.	Make sure that your application software is configured correctly for your printer. See your application documentation. If necessary, install or reinstall the printer driver software as described in the Setup Sheet.
Software settings override settings you make using the default-setting mode or the control panel buttons.	Use your application software to select the font.
Vertical printed lines are not aligned.	
Cause	**What to do**
Bidirectional printing, which is the printer's standard printing mode, is out of alignment.	Use the printer's bidirectional adjustment mode to correct this problem.

The printer does not print when using a network.

Cause: The network setting is wrong.

Solution: Check whether or not print jobs from other computers are being printed through the network. When the print job is executed, the printer or your computer may have a problem. Check the connection between the printer and your computer, settings on the printer driver, and settings of your computer. When the print job is not executed, network settings may be incorrect.

5.3Preventive Maintenance Tools

5.3.1Computer maintenance

Computer maintenance is the practice of keeping computers in a good state of repair. One type of maintenance is preventive maintenance.

<u>Preventive maintenance</u>

The act of a regularly scheduled check of the computer hardware or software to help ensure it continues to operate properly. Fragmentation, hard-disk issues, bad memory, viruses, filled caches, and registry errors can cause computers to act slow or even stop working. We can avoid computer from this problems using preventive maintenance troubleshooting tools and diagnostic softwares. Below is a list of some preventive maintenance steps you can take with your computer or computer hardware.

1. Cleaning your computer hardware.
2. Downloading the latest drivers for your hardware.
3. Downloading the latest updates for your computer software.
4. Verifying you have the latest anti-virus protection updates on your computer.
5. Running disk software utilities such as Defrag and Scandisk on your hard drive.
6. Deleting unused programs or other files on your computer.
7. If you don't turn off your computer it may be good to reboot your computer every few months, unless the computer is a critical computer such as a computer server.

Preventive maintenance can be described as maintenance of equipment or systems before fault occurs. It is a daily maintenance (cleaning, inspection, oiling and re-tightening), design to retain the healthy condition of equipment and prevent failure through the prevention of deterioration, periodic inspection or equipment condition diagnosis, to measure deterioration. It divided in two sub groups:

a. Periodic maintenance (Time based maintenance - TBM)

Time based maintenance consists of periodically inspecting, servicing and cleaning equipment and replacing parts to prevent sudden failure and process problems. Periodic maintenance is Planned based maintenance.

b. Predictive maintenance

This is a method in which the service life of important part is predicted based on inspection or diagnosis, in order to use the parts to the limit of their service life. Predictive maintenance is condition based maintenance. It manages trend values, by measuring and analysing data about

deterioration and employs a surveillance system, designed to monitor conditions through an on-line system.

Troubleshooting Tools

Multimeter

A multimeter is a device used to measure voltage, resistance and current in electronics & electrical equipment It is also used to test continuity between to 2 points to verify if there is any breaks in circuit or line There are two types of multimeter Analog & Digital .

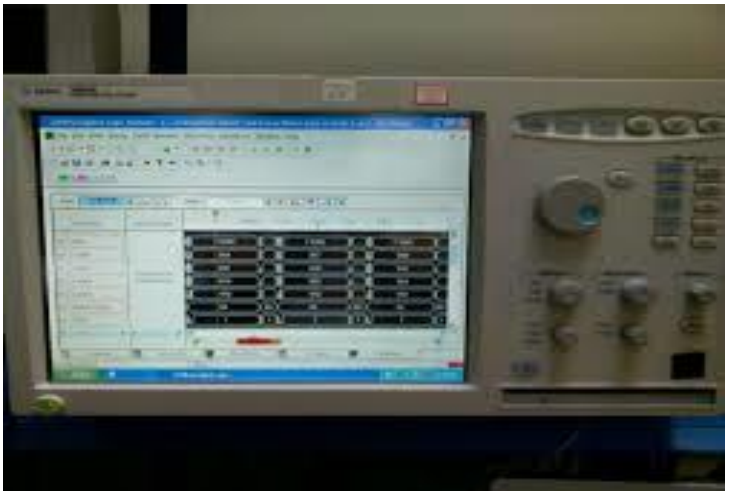
Digital Multimeter

A digital multimeter measures AC / DC voltage, resistance, and current in an electric circuit. It is highly accurate and displays an LCD number readout.

They can be used to troubleshoot electrical problems in a wide array of industrial and household devices such as electronic equipment, motor controls, domestic appliances, power supplies, and wiring systems.

Oscilloscope

An oscilloscope, previously called an oscillograph known as a scope, CRO (for cathode-ray oscilloscope), or DSO (for the more modern digital storage oscilloscope), is a type of electronic test instrument that allows observation of constantly varying signal voltages.



Oscilloscopes are used to observe the change of an electrical signal over time, such that voltage and time describe a shape which is continuously graphed against a calibrated scale. The observed waveform can be analysed for such properties as amplitude, frequency, rise time, time interval, distortion and others.

Oscilloscopes are used in the sciences, medicine, engineering, and telecommunications industry. General-purpose instruments are used for maintenance of electronic equipment and laboratory work.

Logic analyzer

A logic analyzer is an electronic instrument that captures and displays multiple signals from a digital system or digital circuit. A logic analyzer is like an oscilloscope that is used to view multiple digital (binary) waveforms. Logic analyzer use for precise hardware troubleshooting, especially for timing issues. Logic analyzers can be used for any type of digital circuit that has a binary output. Logic analyzers or logic analysers are widely used for testing complex digital or logic circuits.

Logic probe

A logic probe is a hand-held pen-like test probe used for analyzing and troubleshooting the logical states (Boolean 0 or 1) of a digital circuit. There are usually three differently-colored LEDs on the probe's body: Red and green LEDs indicate high and low states respectively and an amber LED indicates a pulse .The pulse-detecting electronics usually has a pulse-stretcher circuit so that even very short pulses become visible on the amber LED. A control on the logic probe allows either the capture or storage of a single event or continuous running. When many logic levels need to be observed or recorded simultaneously, a logic analyzer is used.

Logic Pulser

The logic pulser is hand held tool used to inject pulses at the input of a gate under test. It maps single or stream of pulses at different frequencies.

Current Tracer

The current tracer is hand held tool which detects current flow in electronics circuits. It is used to locate shorted components, track shorts, solder bridge, ground short etc.

<u>Diagnostic Software</u>

- Diagnostic software is used to identify problems on a computer or piece of equipment.
- These programs test the onboard systems for issues and help to alert users of potential problems or breakdowns.
- They often provide hints or at least valuable information about what is in the system and how it is working.

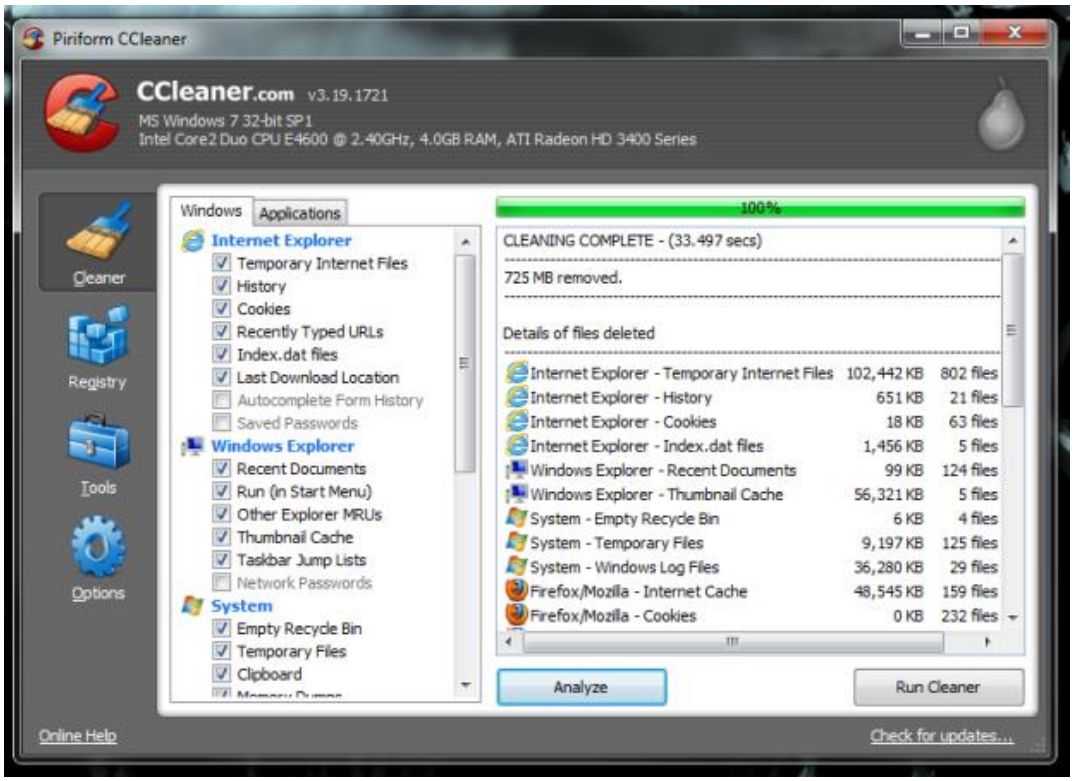
1: MemTest

MemTest is a simple test that determines whether your computer can reliably store information in its RAM. With this tool, you can define how much memory to test and then run the test (with

all applications closed). If an error is reported, your RAM is suspect. The only drawback to MemTest (and most software memory testers) is that it can't pinpoint which memory chip is bad. But if you're looking for a memory tester that can be run while Windows itself is running, MemTest is your tool.

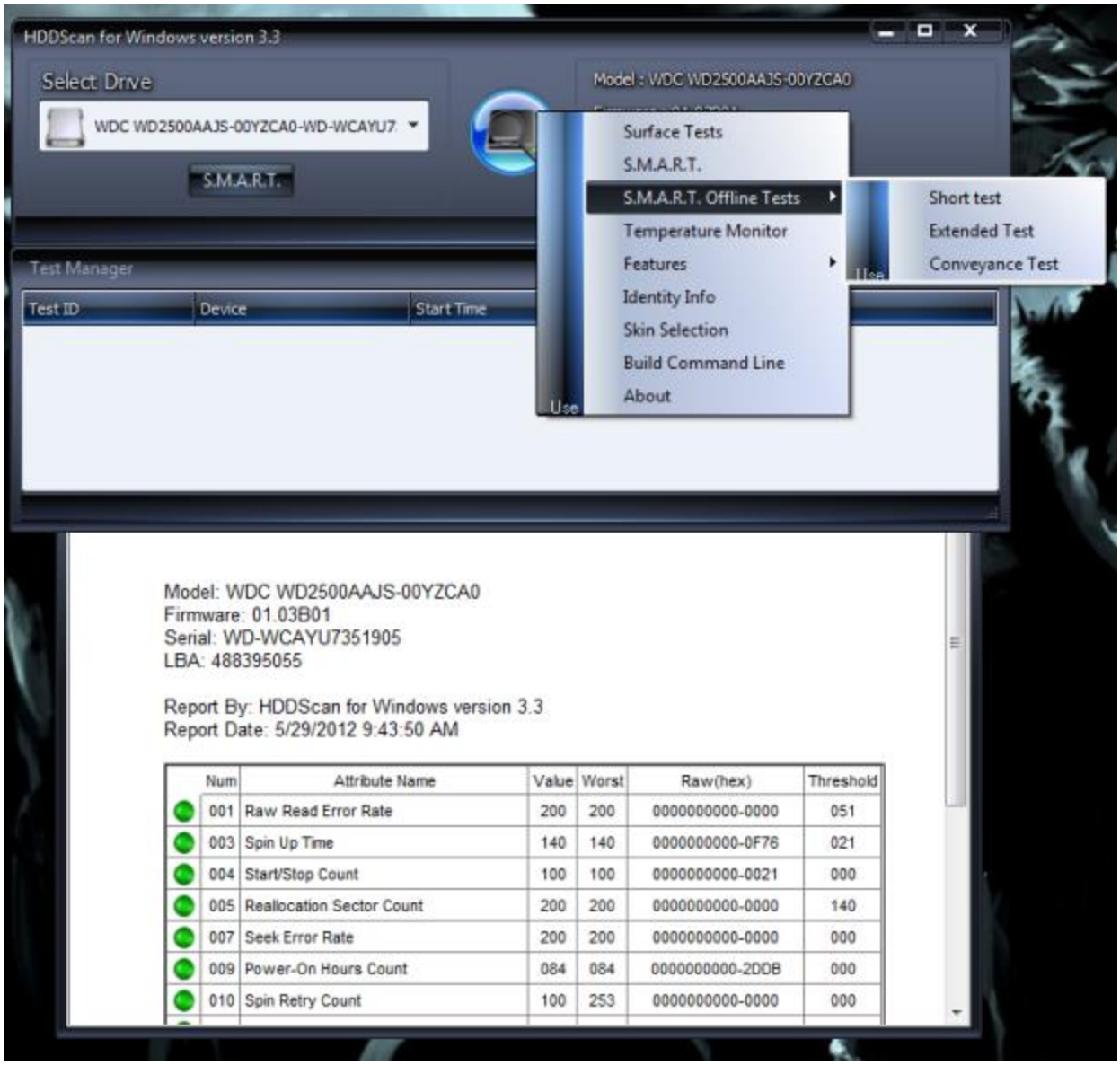
2: CCleaner

CCleaner can help you clean out of your machine. Not only can CCleaner keep your disk cache clean, it can keep your registry free of errors. The result in a computer running efficiently.Using CCleaner regularly will go a long way toward keeping your machines running well.



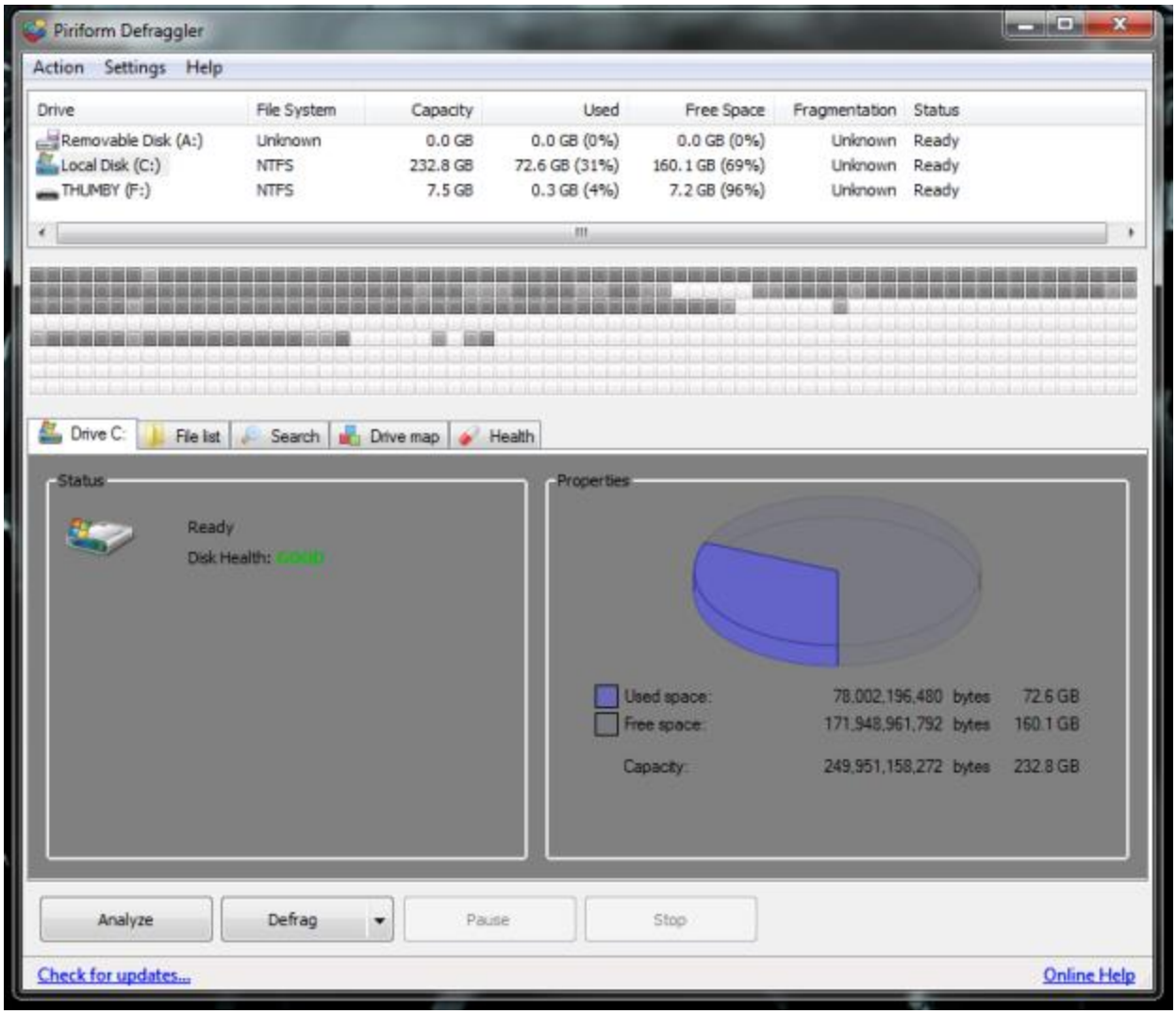
3: HDD Scan

HDD Scan is a free hard disk diagnostic tool that supports standard drives, RAID arrays, Flash USB, and SSD drives. With this tool, you can scan for errors (bad blocks and bad sectors). By regularly using this tool, you may avoid a avoid disaster.



4: Defraggler

Defraggler is one of the best defrag programs available. It will defrag an entire hard disk as well as individual files or folders. You can specify what you want to defrag and how you want to defrag it. Defraggler also offers a portable version, so you don't have to install the tool on every machine you touch. Defraggler setting up regularly scheduled defrags to keep every machine on your network running smoothly.



Dr. Watson (debugger)

- Dr. Watson is an application debugger included with the Microsoft Windows operating system.
- It may be named drwatson.exe, drwtsn32.exe or dwwin.exe, depending on the version of Windows.
- it would collect error information (symptoms) following a program crash.

- The information obtained and logged by Dr. Watson is the information needed by technical support personnel to diagnose a program error for a computer running Windows.

- A text file (usually drwtsn32.log) is created whenever an error is detected, and can be delivered to support personnel by the method they prefer.